

Necromancing Diels: computerising the phonological analysis of early Slavonic texts using existing treebank data and a Late Common Slavonic computerised inflectional morphology

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ABSTRACT

This article demonstrates a new way of massively speeding up traditional historical phonological investigations into early Slavonic philological material by furnishing each word of a manuscript-text with its direct Late Common Slavonic (LCS) reconstructed ancestor. Such reconstructions act as a type of "phonological annotation", or index, since they make the reflexes of any LCS phoneme-sequence reliably and predictably searchable, regardless of a manuscript's surface-spellings. To automate the production of this LCS reconstructed layer, a comprehensive computerised LCS inflectional morphology has been developed, which is capable of using existing lemmatisation and morphology-tagging annotation-data from texts in the Tromsø Old Russian and Old Church Slavonic (TOROT) corpus in order to inflect LCS lemmas according to each text-word's morphological shape. A web-interface, ocstexts.co.uk, is provided to showcase some of the benefits of the resulting "phonological annotations", including searchable autoreconstructed versions of OCS texts from the TOROT corpus, with virtually 100% coverage of the Codex Marianus, Codex Zographensis, and Psalterium Sinaiticum.

0. Introduction

Much progress has been made in the last twenty years in early Slavonic corpus linguistics as a result of the Old Church Slavonic part of the PROIEL project (Haug & Jøhndal 2008) and its subsequent expansion as the TOROT treebank (Eckhoff & Berdičevskis 2015), such that currently just over 240,000 words of canonical OCS have been manually lemmatised, part-of-speech and morphologically-tagged, and syntactically parsed. The focus of these projects, however, has been exclusively on the higher-level linguistic domains of syntax, semantics, and pragmatics: surface-morphology has been of only incidental concern, for example in investigations into differential-object marking (Eckhoff 2015, 2022), but the sort of inflection-class-marking which would enable the retrieval of, say, masculine *o-stems vs *i-stems is lacking. The needs of historical phonologists especially are ill-served, since some of the texts contain quite severe typographical inconsistencies and errors that make them impossible to search reliably and dangerous to use without reference to the manuscripts.

That being said, enough information is included in TOROT's lemmatisation and morphology-tagging that, with a few exceptions (e.g. certain comparatives), the morphological shape of the inflected text-forms can be predicted from just the tag-information, provided that inflection-class annotations are added to the lemmas. This means that the immediate Late Common Slavonic ancestors of surface-text forms can be generated by reconstructing and inflection-class-marking the LCS lemmas, and then using a computerised LCS inflectional-morphology to inflect each text-word's lemma according to its morphology-tag¹. Such LCS reconstructions are an extremely useful form of 'phonological annotation', because theoretically all the information required to give rise to an attested form must by definition be present in any correct reconstructed proto-form, and the complete regularity of the idealised LCS forms makes texts predictably searchable regardless of orthographic variability, abbreviations, or other irregularities in the surface-texts. When applied to whole texts, they make the exhaustive investigation of almost any phonological or orthographic question trivially easy compared to manually reading and extracting relevant forms, or using

1 Morphological innovations and variations are detected by inspecting the text-forms and then applying 'alternative' endings as specified in the inflectional-endings database; see Section 2.3.

TOROT's existing lemmatisation and morphology-tagging to try to gather morphological categories which might contain the sound-groups one is interested in.

The goal of this article is to describe just such a computerised LCS inflectional-morphology, and to show how it can be used to "autoreconstruct" different OCS texts from the TOROT corpus, while explaining how difficulties caused by things like morphological innovations, badly-integrated foreign loanwords, or insufficiently-precise tagging-data can be overcome. The resulting 'phonologically annotated' texts should allow investigators of the lower-level linguistic domains (phonology, orthography, and morphology) to benefit from the huge amount of manual annotation work that has gone into the OCS part of TOROT, and the computerised LCS inflectional-morphology by itself is valuable in that it allows linguistically-rigorous and comprehensive inflection and conjugation-tables to be generated².

A web-interface for autoreconstructed TOROT OCS texts is being developed at <https://ocstexts.co.uk>, which aims to demonstrate the benefits of this extra layer of annotation by offering things like LCS phoneme-search of texts, conversion of each autoreconstructed word into canonical "normalised" OCS, for easy comparison with the manuscript-form, and computer-generated inflection and conjugation-tables for every reconstructed lemma. It also includes TOROT's existing morphology, lemmatisation, and (for three of the texts) Greek source-alignment data, plus integration of the recently digitised OCS dictionaries hosted at gorazd.org.

The usefulness of computational methods for diachronic phonology has of course not escaped the notice of previous scholars, and Sims-Williams (2018) offers a useful survey of a field he calls "Mechanising Historical Phonology". Of the three main strands of work considered there, ("automatic cognate-recognition", "computerised forwards-reconstruction", and "computerised backwards-reconstruction"), only forwards-reconstruction is of any relevance to the task of holistically investigating the sound-systems of manuscripts, in that one could apply sound-changes and orthographic rules to the LCS reconstructed layer of individual texts to try to "reverse engineer" the manuscript spellings, and in so doing test hypotheses about the phonological features of the underlying dialect[s] that gave rise to them. The forwards-reconstruction work mentioned by Sims-Williams, though, is invariably aimed at producing sets of isolated, regularised forms, either from modern standard languages, or theoretical idealised forms from old languages, e.g. Borin's (1988) OCSGen system that tried to produce "normalised" OCS from PIE. This type of work is useful for identifying problems with traditional diachronic accounts of ordered sound-rules³, but it is of a fundamentally different nature to what I'm doing here, which is about elucidating the concrete philological facts of the manuscripts in order to better understand the synchronic linguistic systems responsible for them.

Most early Slavic texts are from dialect-areas (East Slavic and Bulgaro-Macedonian) whose modern phonological systems show evidence of sharp differentiation along deep structural lines, most importantly in the phonologisation of secondary consonant-palatalisation and the tolerance for bimoraic syllables (i.e. distinctive vowel-length). The seeds of these differences must have been

2 The lemma and inflection-class data included as a supplement to Polivanova's (2013) comprehensive OCS grammar, <https://integral.github.io/osd/>, would in principle allow similar tables to be generated, but as far as I know Polivanova has not digitised the actual endings, nor written any scripts to generate paradigms. Additionally, because she bases her forms on a normalised OCS, rather than LCS, phonemic system, a massive amount of information has been needlessly destroyed. To give just one obvious example, the syllabic-liquids *ř *ľ *ṛ *ḷ are represented by their OCS reflexes /рь лъ рѣ ѣ/, meaning there's no distinction between the syllable-rimes of roots like /-ръьь-/ <*řd and /-ръььт-/ <*krst. An inflectional-morphology based on her data would therefore only be good for generating the (in reality unattested and fictional) "normalised" form of OCS, whereas my LCS forms are trivially convertible into any dialect of OCS, as well as pre-Jer Shift Old East Slavic (see pg. 17 and Figure 6), because they represent the language at a stage before any such dialect-specific phonemic mergers.

3 For a recent example see Marr & Mortensen (2020), who found many problems with the traditional account of how Latin became French, and were able to improve upon it using their suite of software for performing forwards-reconstruction and "debugging" problematic sets of rules.

sown around the time of the radical restructuring occasioned by loss of the reduced-vowels * $\bar{u}$, $\bar{e}$ (the so-called "Jer Shift"), i.e. the very time at or shortly after which our manuscripts were written, but the limitations of the original writing-system (which was devised for a language which had *not* yet undergone the Jer Shift) mean that they do not find clear expression in the texts and can only be dimly groped at through indirect means. Much more thorough investigation of *all* aspects of the manuscript-spellings are thus required in order to uncover the nature of the linguistic systems they reflect.

1. My Late Common Slavonic (LCS) phoneme-system

In order to account for as much of the subsequently attested Slavic as possible, a point after the monophthongisation of diphthongs, but before the Second and Third Velar Palatalisations (PV2 and PV3) is chosen as the point of departure, because of the difference between the West Slavic /š/ and South/East /ś/ reflex of these two palatalisations of *x (Cz. loc. pl. *dušič* vs Suprasliensis *доуѣхъ* <*duxěxъ; Polish *wszak* vs Supr. *вѣсакъ*, Ru. *всѣк[уй]* <*vьx-akъ), as well as the probable complete absence of PV2 in northern East Slavic⁴ (Old Novgorodian, see Zaliznjak 2004: 42-45 for

the evidence), and the blocking of PV2 by an intervening *v in West Slavic (Pol. *gwiazda*, Cz. *květ* <*gvězda, *kvěť, etc.).

To be explicit, the native phonemes in my LCS system are given in the tables below:

Table 1: LCS Vowels after the monophthongisation of diphthongs

	Front		Back	
High	i		y	u
	í ь Ї		у ҃ ы І	
Mid	e ę		o ɔ	
	ě ǣ			
Low	ā			
		a		

Table 2: LCS consonants before PV2/PV3 (adapted from Winslow 2022: 304)

Labial	Dental	Palatal	Velar
m	n	ń	
b p	t d	h ħ	k g
	s z	š ž	x
		č	
	l	ĺ	
	r	ř	
v		j	

1.1 Foreign sounds

In addition, the following symbols are used for phonemes of wholly foreign origin in order to represent badly-integrated foreign borrowings, whose level of integration into the native system we deliberately do not take a position on: /k ɡ ʃ x f ü/, e.g. in respectively *кѣтъ* <*kítъ, *ѣмѣмонъ* <*igemonъ, *хитѣмонъ* <*xítomъ, *иосифъ* <*ijosifъ⁵, and *мѣро* <*müro. Almost none of the words containing these symbols would actually have existed in the language during Common Slavonic times, but they need to be included in the indexing-system because they often contain native Slavic elements (f.ex. inflectional endings). Normally they represent specific sounds in the source-language (usually Greek), so including them is useful for investigating the process of these sounds' integration into the native systems. For instance, the extent to which Greek /ü/ is integrated into either native /i/ or /u/ can be seen in variations in the OCS spellings of the word for 'Egypt' (*egüptъ): *ѣгѣпѣтъ* {egüpta} vs *ѣгѣпѣтъ* {egipta} vs *ѣгѣпѣтъ* {egüptü} vs *ѣгѣпѣтъ* vs *ѣгѣпѣтъ* {egüpta}⁶, etc.. One might also ask whether a separate <ѣ> letter for /ǵ/ (and the writing of <ѣ>/<ѣ>

More importantly, as Галинская (op. cit.) points out, in all of the well-known Novgorodian forms of the pronoun *vъxъ 'all' which supposedly show a lack of PV3 by retaining both /x/ and back/hard desinences (e.g. fem. gen. sg. *вѣхѣ* <*vъxoĭĕ from letter №850), and which come from letters which otherwise correctly convey the jers (by writing <ѣ, ѣ> for *ъ and <ѣ, ѣ> for *ъ), the weak-jer is always written with <ѣ, ѣ>, unambiguously suggesting a /ъ/ pronunciation. These forms therefore more likely point to a LCS doublet-form *vъxъ which would never contain the conditioning environment for PV3 anyway, and thus you can't use them as evidence of a lack of PV3 in Novgorodian (on the plausibility of such a doublet see Галинская (2014: 14), though cf. Zaliznjak's (2004: 54) less convincing explanation of the /ъ/ in these words as an assimilation of original /ъ/ to the back-vowels of the following syllable).

5 Of course the sequence /jo/ violates LCS phonotactics as well.

6 Forms are given as they appear in the manuscripts; modern fonts and Unicode symbols mean that the misleading and unhelpful practice of transcribing Glagolitic into Cyrillic is no longer defensible. Latin-based transliterations of Glagolitic forms are given in {curly braces} according to the table in Appendix 1, but it should be emphasised that no transliteration system can adequately convey the internal logic of the Glagolitic, and those who study OCS through transliterations are hamstringing themselves. Where specific forms from Eckhoff's corpus-texts are cited, they are hyperlinked to the corresponding part of the text as it appears on the ocstexts.co.uk website. The Glagolitic texts there are unfortunately still Cyrillicised and should be used with care, particularly ones like Psalterium Sinaiticum where certain ruesome decisions from the editors Severjanov (1922) and Mareš (1997) have been compounded by further information-destroying simplifications (for instance, Severjanov transliterates <ѣ> with <ѣ>, but Eckhoff then replaces Severjanov's roundy diacritic with a tilde, leading to nonsense transcriptions like Psalm 8 <ѣѣѣѣ> for ms. <ѣѣѣѣ>). Cleaned up digitisations, using the Glagolitic originals as the base-text, and a

1.2 Vowels

The two extra nasal-vowels /ɣ/ and /ǣ/ are required to account for the split between North (East and West) and South Slavic forms of certain inflectional-endings:

- The need for the retention of the / $\bar{\text{E}}$ / archiphoneme, which represents merged Early Common Slavonic * $\bar{\text{e}}$ * $\bar{\text{a}}$ in the position after palatal consonants, up to this point of LCS, is explored in detail in Winslow (2022), but the same archiphoneme (along with its short counterpart / E /) was explicitly posited by Kortlandt as far back as 1979 (p.266) as part of his ECS system. In short, a combination of:

mechanism for displaying manuscript-images, are planned for the near future.

7 The other troublesome pre-LCS morphological isoglosses reflected in the texts include the masc./nt. instr. sg. *o- and *jo-stem endings *-ьмъ/*-ьмь (N.Sl., e.g. KF **ѡбразѹмѣ** {obrazŭmĭ}, Uspensk. Sbor. **КНАЗЪМЬ**) and *-омъ/*-емь (S.Sl., e.g. Supr. **ѡБРАЗУМЪ, КНАЗЕМЪ**), which are most commonly (e.g. Olander 2015: 168) thought to be analogical replacements of the original instr. sg. ending ECS *-ā which is preserved in the adverb *vŷčera ‘yesterday’; and the *-тъ (N.Sl.) vs *-тѣ (S.Sl.) verbal endings of 3rd sg. and pl. present (plus its extension to 2nd and 3rd sg. aorists like OCS **НАЧАТЪ, ОР** (Uspensk. Sbor.) **БЫСТЪ, НАЧАТЬ**). Here I have no choice but to index them with dummy-symbols in the database: *-Омъ/*-Емь for the instr. sg. ending and *-tQ for the verb-endings.

2.) overwhelming spellings of palatal-letter (<ѡѡѡѡ>) + jat' in the Kiev Folia (the oldest and therefore least distant ms. from the 'original' OCS, as first codified by Cyril and Methodius and for which the Glagolitic alphabet was devised) for the reflexes of LCS *č, *š, *ž, *h + *Ē (e.g. ѡѡѡѡѡѡ {oběčlŭ} <*ob-věhĒlŭ, ѡѡѡѡѡѡ {dušēmŭ} <*dušĒmŭ), as well as occasional traces of such spellings in later Glagolitic OCS (e.g. Psal. ѡѡѡѡѡ {čěšę} <*čĒšę); and

3.) the evidence of certain modern Bulgarian dialects, which have reflexes of LCS *ě in words like *ѡѡѡѡ* <*žĒba 'toad' (Stojkov 1954: 74–78),

all together point very strongly towards there having occurred a split on the Southeastern periphery of Slavic between LCS dialects which have /ě/ < *Ē and the majority of the rest which got /a/, and that original OCS ('Urkirchenslavisch') was an *Ē > /ě/ dialect. I posit that *Ē remained until the opposition /a:/ /ě/ after palatal consonants was reintroduced when PV2 and PV3 brought new soft-consonants /c š dž/ into the system, which could be followed by both /a/ and /ě/: fem. nom. sg. /stědža/ < *stĕga (PV3) vs fem. loc. sg. /nodžě/ < *nogĕ (PV2) (Winslow 2022: 304-305). Thus since my LCS system is based on a point just *before* PV2 and PV3, I must also retain the *Ē archiphoneme.⁸

The syllabic liquids /r̥ r̥ l̥/ are included as unitary vocalic phonemes, following Schenker (1995: 94), rather than as combinations of /r̥ l̥/ + /r̥ l̥/, because these groups descend from PIE syllabic liquids and many descendant South Slavic dialects which retain syllabic liquids in this position (including most of those underlying canonical OCS) do not show any evidence of an intervening oral-vowel + liquid stage (such a view is shared by Bethin (1998: 71-72); cf. also Bulgarian dialectal evidence in Stojkov (1954: 130-131), where hard consonants precede reflexes of the LCS /r̥ r̥/ even in dialects with secondarily-palatalised consonants before fallen weak LCS /r̥/). The need for both front and back *r̥ *l̥ is unambiguously shown by the East Slavic reflexes /er/ and /or/ (Ru. *смерть*, *морковь*), but *r̥ vs *l̥ is more complicated: PIE *p̥l̥nos, *w̥l̥k̥os > Lithuanian *pilnas* 'full', *wilkas* 'wolf' (LCS *p̥l̥n̥s, *v̥l̥k̥s) vs Lith. *stulpas* (LCS *stl̥p̥s 'pillar') suggests that Balto-Slavic had differentiated front/back variants of the PIE syllabic *l̥ (Bethin 1998: 69), but the ancestor to East Slavic backed all vowels preceding tautosyllabic /l̥/ (Ru. *молоко* < Proto-ESl. *molko < LCS *melko > OCS *млѣко*), and thus only has /ol/ reflexes here: Ru. *волк*, *столб*, *полный*. It's true that Polish has *wilk* and *milczeć* (<*m̥l̥čĒti), but the Polish reflexes are complicated and likely have more to do with the surrounding consonants: *p̥l̥n̥s by contrast gives *pełny* with hardened /l̥/ and the Polish non-palatalising-/e/ reflex of *r̥, and the differing reflexes in *wierzch* <*v̥r̥ch, *śmierć* <*s̥m̥r̥t̥ and *martwy* <*m̥r̥t̥v̥ rule out any explanation based on the nature of the LCS syllabic-liquid alone (for more discussion see Bethin op. cit.: 73-75). While most OCS shows no sign at all of a front-back distinction in the syllabic-liquids and writes the reflexes of these groups overwhelmingly with <ѣ> and <лѣ>, the Kiev Folia, which is the only OCS text that reflects a pre-Jer Shift stage and is very nearly flawless in its etymologically correct rendering of the jers, also spells *r̥ *l̥ and *r̥ as one would expect: ѡѡѡѡѡѡ {skvr̥n-} ѡѡѡѡѡѡ-

8 It's possible to argue that the short *Ē counterpart to *Ē persisted in East Slavic until after the Fall of the Jers, and that the ES. so-called e > o shift before hard-consonants / back-vowelled syllables is actually just the resolution of this archiphoneme as /o/ (where palatalisation of the preceding consonant remained, in e.g. Ukr. *бджола* <*bĕčĒla, or was newly phonemicised, in e.g. Ru. *вѣсла* <*vĒsla <*vesla), and that there was never a stage when these words had /e/ (based among other things on <o> spellings regardless of stress after palatal-letters in very early texts, and even after the letters for secondarily-soft LCS plain consonants in the Birchbark documents (Le Feuvre 1993, Nakonečnyj 1962), but there isn't space to elaborate on the issue here (see Winslow 2022: 304 fn.16). Unlike the situation with long *Ē, OCS shows no sign of anything but an /e/ reflex of short *Ē (and indeed the fact that the East Slavs inherited their writing system ultimately from the Urkirchenslavisch system designed for such a dialect, rather than one which had a clear way of writing /soft consonant/ + /o/, is likely the reason that /o/ reflexes are so rarely detectable in the early texts, since <e> had to be used for both /e/ and /'o/, cf. the spelling ѡѡѡѡѡѡ of the Kipchak word /jovšan/ 'wormwood' in the Hypatian Codex, whose modern cognates (Turkmen *yowšan* /jowšan/, Kazakh *жусан* /žuwsan/, Azeri *yovšan*) unambiguously point to a Kipchak /o/), and the history of the East Slavic /o/ reflexes remains the subject of much disagreement, so it's simpler for everyone if I continue the traditional practice of writing LCS *e after palatals, even if that strictly speaking is inconsistent with my use of *Ē.

{tvrīd-} 𐌹𐌴𐌹𐌳- {srīdic-} 𐌱𐌹𐌳𐌺- {driž-} <*r̥, 𐌸𐌹𐌵𐌹𐌳𐌹𐌴 {skrūbīni} <*r̥, and 𐌱𐌹𐌳𐌹𐌴𐌹𐌴 {naplīneni} <*l̥ (Winslow 2022: 313), and even Zographensis spells all 5 occurrences of *vľk- 'wolf' with 𐌹𐌵𐌹𐌴-/𐌹𐌵𐌹𐌴- {vľk-/vľc-} and all 15 instances of its *-mlĭč- root with -𐌹𐌵𐌹𐌴- {-mlĭč-} (e.g. 𐌹𐌵𐌹𐌴𐌹𐌴𐌹𐌴 {umlīčašn}). Therefore, taken as a whole the Slavic evidence pretty securely points to front and back variants of both syllabic liquids, and for searching purposes it's far preferable to denote them with separate symbols⁹ rather than as the sequences /ɛr ɛr ɛl ɛl/¹⁰.

1.3 Consonants - Dejotation

Reflexes of the so-called jot-palatalisation are all written either as unitary palatal phonemes, or in the case of jot-palatalised labials as /vľ mľ bľ pľ/, rather than as sequences of consonant + /j/, hence /ń ľ ř/ for *nj *lj *rj. The 'dejotated' reflexes of *tj (and *kt+front-vowel) and *dj are denoted using the modern Serbian Cyrillic letters /h/ and /ħ/ respectively, because the commonly used alternatives, i.e. /t̥ d̥/ (as used in e.g. Olander 2015) or /k̥ ġ/ (as used in Winslow 2022), or variations thereof, are visually too close to symbols used elsewhere in the system. /k̥, ġ/ are anyway already used in my system for foreign /k, g/ before front-vowels, and /t̥ d̥/ look too similar to the common denotations of secondarily-palatalised post-Jer Shift /t' d'/, as used in discussions of systems like Russian or Eastern Bulgarian where they arise.

The compelling hypothesis, first proposed by Durnovo (1929: 55-58) but most recently elaborated by Vermeer (2014: 209-214), and accepted by Mathiesen (2014: 197 fn. 22) and Winslow (2022: 310 fn.25), according to which the Urkirchenslavisch reflexes of *h, ħ were close enough to foreign /g k/ before front-vowels that the original Glagolitic system used <ꙗ > for both sets of sounds (i.e. alongside attested 𐌸𐌹𐌵𐌹𐌴𐌹𐌴 {igemonŭ} < ἰγεμών would have been **𐌸𐌹𐌵𐌹𐌴𐌹𐌴 {osogēni} <*osohēni, and alongside attested 𐌸𐌹𐌵𐌹𐌴𐌹𐌴 {dŭčerĭ} <*dĕħerĭ would have been **𐌸𐌹𐌵𐌹𐌴𐌹𐌴 {ċinŭsŭ} < κῆνος¹¹), does not prevent us from keeping the foreign sounds separate for our LCS stage, since clearly they differed enough in all the dialects underlying actually attested OCS to be written separately.

Pre-dejotation *stj and *zdj are differentiated from the PV1 reflexes of *sk and *zg by writing the former as *šh and *žh and the latter as *šč and *žž, even though their modern reflexes do not differ from each other anywhere and so must've fallen together in the CS period, because they often alternate with their respective un-palatalised counterparts morphologically and derivationally, e.g. očistiti:očišhensje vs. jьskati:jьščq, jĕzditi:jĕžhŭ vs jьzgъnati:jьžženq.

There are convincing arguments for PV2/3 having preceded dejotation, at least in more central areas, most recently presented in e.g. Vermeer (2014: 197) and Wandl & Kavitskaya (2023: 244-

9 In the database I will have to use the single Unicode characters <ř ľ l̥>, rather than what's shown in Table 1, since the latter cannot actually be rendered without using the letters for /r ľ l̥/ plus the 'combining ring below' U+0325 symbol, which means searches for the consonantal liquids on their own will also return results containing syllabic liquids. A similar problem affects /ċ y/, which I will have to replace with <ċ ŷ>.

10 To my mind the only evidence in support of a genuine jer + liquid stage comes from the paradigms of verbs like OCS 𐌸𐌹𐌵𐌹𐌴𐌹𐌴 <*sŭtŭti, where the syllabic /ř/ in the stem alternates with /ɛr/ depending on the vocalicity of the following morpheme: the e.g. 3sg. pres. *sŭtŭretŭ (Zogr., Supr. 𐌸𐌹𐌵𐌹𐌴𐌹𐌴) or (one possibility of the) 3rd sg. aorist *sŭtŭre (Supr. 𐌸𐌹𐌵𐌹𐌴) must have /ɛre/, while the 3rd pl. aorist *sŭtŭřę (Supr. 𐌸𐌹𐌵𐌹𐌴) and the other possibility for the 3rd sg. aorist *sŭtŭ (Psal. 𐌸𐌹𐌵𐌹𐌴 {sŭtrĭ}, or with a different prefix Mar. 𐌸𐌹𐌵𐌹𐌴 {otrŭ} <*otrĭ), being word-final or pre-consonantal, must be syllabic /ř/. The same alternation occurs in the zero-grade forms of verbs like *umerti, as is clear from the Polish reflexes *umarł* <*umřĭ vs *umrę* <*umrĭ. The argument could be that at some stage, before the LCS tendency towards Open Syllables became dominant, the stems in these paradigms were surely unitary /tŭr/, /mŭr/, i.e. 3sg. aor. /sŭ.tŭ.re/ vs 3rd pl. aor. /sŭ.tŭr.řę/, and that the latter's closed /tŭr/ syllable was only forced to open itself up by changing to /tŭř/ because of the Law of Open Syllables. Thus at least one source of the syllabic-liquids could be shown to have developed from a vowel + liquid stage, but that still doesn't prove that they all did, or that the change of /tŭr/ to /tŭř/ in these verb-forms was not merely a move to an already-existing syllabic-liquid phoneme.

11 Interestingly, this aspect of the hypothesised Urksl. orthographic system has rearisen in the modern Macedonian standard due to Turkish loanwords: *кемер* < Tk. *kemer* 'belt', *ке* < *[x̥]he[t̚]; *ѓон* < Tk. *gön* 'leather', *меѓу* <*meħu.

247), and therefore it could be objected that my system, which contains the dejotation reflexes /h ħ ŋ í ř/ but not the PV2/3 reflexes /c ś dž/, is ahistorical. However, it should be reemphasised that the primary goal of my LCS reconstructions is to act as an index which allows reflexes in texts to be found, not to be a historically realistic description of some actually-existing LCS dialect. The absence of PV2 in Novgorodian shows that it can't have preceded dejotation everywhere in Slavic, and in any case the replacement of the sequences /tj dj nj lj rj/ by articulatorily distinct combined units, no longer associated by speakers with their /t/ and /j/ phonemes, is structurally completely irrelevant unless and until these new units merge with existing phonemes (or new sequences of dental + /j/ are introduced), as e.g. in the KF dialect where /tj/ merged with /c/ from PV2/3, or in ESL where it merged with /č/ from PV1. A language which had distinct Czech-like palatal [c, ʃ] reflexes of *tj and *dj, and also no new sequences of [tj, dj], could not convincingly be argued to have undergone dejotation at the phonemic level, as these new units would just be phonetic realisations of /tj, dj/. Analysed like that, the symbols /h ħ ŋ í ř/ in my system strictly speaking would really just be cover-symbols for the pre-jotation sequences, but such notation is preferable since it prevents searches for groups containing /j/ alone from returning results polluted by all the dejotation-groups. As explored in Winslow (2022), the status of /j/ as a phoneme in the earliest OCS texts is an intricate problem, so the ability to investigate the reflexes of *j in isolation from the dejotation-reflexes is important.

1.4 Issues involving the glide /j/

1.4.1 Word-initial *jĀ-/ *a-

The tendency for ECS *ā- to have taken prothetic /j/ by LCS times (in accordance with the drive towards open syllables) can make it difficult to distinguish this group from *jĀ- in the absence of wider Indo-European evidence. Normally I've followed Derksen (2008), or the ESSJa (*Этимологический словарь славянских языков*, Trubačev 1974-), but for certain lexemes, e.g. *ama 'pit', which in OCS is spelt overwhelmingly with ⱭⱮ- {ēm-} or Ɑм-, the single Greek cognate ἄμη adduced by ESSJa I p.70 in favour of jot-less *am- is not enough to categorically exclude the alternative *jĀema. In particular the 1sg. nom. pronoun *azъ/*jĀezъ is especially problematic: I follow ESSJa I p.100 which ultimately plumps for *azъ, but Derksen doesn't discuss it at all. (A lengthy discussion of the evidence can be found in Teneva's (2012) article on the subject.)

Forms with insecure etymologies can't under any methodology be used as good evidence in phonological investigations, so in difficult cases like the above I simply mark the lemma in the database and provide some short discussion, so that eventually the web-interface can flag such forms in some way and inform users of the specific difficulties.

Like Derksen, I assume that roots going back to PIE jot-less long *ē or diphthongal *oi-, e.g. the root for 'to eat', PIE *h₁ēd-, all took prothetic *j and merged with *jĀ- from other sources, unlike Durnovo (1929: 54), who seems to think that such a development was limited to Bulgarian and Macedonian dialects, including those underlying OCS (where in the Cyrillic mss. we get regular Ɑѣти etc.). Isolated nominal forms like Ru. *язва* (which Derksen (2008: 155) derives from a Balto-Slavic *oi- based on Lith. *aiža* and Old Prussian *eyswo*) suggest that *ě reflexes in the modern forms of verbs like Ru. *есть*, Pol. *jeść* are later generalisations from prefixed forms like OR Ɑѣтъ, where no jot-prothesis could take place (cf. Schenker 1995: 88, Winslow 2022: 302 fn.14).

1.4.2 Jers before *j

As explored more fully in Winslow (2022: 313-315), OCS spellings seem to suggest that free-variation between <ѣ, ѣ> and <и, ѣ> was a feature of the pre-Jer Shift Urkirchenslavisch orthographic system for conveying the reflexes of the sound-groups *ĭj and *ъj (so-called 'tense jers'), regardless of whether they were in strong or weak position. The examples given were: Zogr.

𐌱𐌹𐌿𐌸𐌰𐌹𐌸𐌰𐌹𐌸𐌰 {znamenīĕ} vs 𐌱𐌹𐌿𐌸𐌰𐌹𐌸𐌰𐌹𐌸𐌰 {znamenīĕ} <*znamenĭjĕ, 𐌳𐌹𐌿𐌸𐌰𐌹𐌸𐌰 {udarīi} <*udarĭjъ vs 𐌳𐌹𐌿𐌸𐌰𐌹𐌸𐌰 {omočii} <*omočĭjъ; Mar. 𐌲𐌹𐌿𐌸𐌰𐌹𐌸𐌰𐌹𐌸𐌰 𐌱 {osqđntīi} vs 𐌲𐌹𐌿𐌸𐌰𐌹𐌸𐌰𐌹𐌸𐌰 𐌸 {osqđntīi} i <*osqđetъ ѱъ; KF 𐌲𐌹𐌿𐌸𐌰𐌹𐌸𐌰𐌹𐌸𐌰 {milostīĭ} vs 𐌲𐌹𐌿𐌸𐌰𐌹𐌸𐌰𐌹𐌸𐌰 {milostīĭ} <*milostĭjŏ, 𐌶𐌹𐌿𐌸𐌰𐌹𐌸𐌰𐌹𐌸𐌰 {vīsemogīi} vs 𐌶𐌹𐌿𐌸𐌰𐌹𐌸𐌰𐌹𐌸𐌰 {vīsemogīi} <*vĭxomogyĭjъ). Of importance here is the fact that the same orthographic system characterises both pre- (i.e. KF) and post-Jer Shift texts; that even in strong position in a text like Zographensis, which shows pretty clear signs of having undergone the Jer Shift, spellings like 𐌳𐌹𐌿𐌸𐌰𐌹𐌸𐌰 {udarīi}, 𐌶𐌹𐌿𐌸𐌰𐌹𐌸𐌰 {boīi}, 𐌶𐌹𐌿𐌸𐌰𐌹𐌸𐌰 {vnštīi} for what in the live dialect underlying Zogr. must surely have been /udarĭj/, /boĭĭj/, /vęštĭj/, are not infrequent¹². The fact that the same alternation occurs in the pre-Jer Shift KF (i-stem gen. plurals 𐌲𐌹𐌿𐌸𐌰𐌹𐌸𐌰𐌹𐌸𐌰 {zapovĕdīi} <*zapovĕdĭjъ vs 𐌲𐌹𐌿𐌸𐌰𐌹𐌸𐌰 {lūdīi} <*lūdĭjъ) suggests that it is a common inheritance from the Urkirchenslavisch spelling system, and thus that in pre-Jer Shift Slavic the difference between /ъ ѱ/ and /i y/ was neutralised before /j/, and we should perhaps posit archiphonemes (which I call /Ī/ and /Ŷ/) in this position. These archiphonemes are, in slightly different terms, effectively posited by Trubetzkoy (1954: 70) in his analysis of the Urkirchenslavisch phoneme-system¹³.

Difficulties arise though when deciding how to denote foreign sources of /ij/¹⁴ which may or may not have been integrated into the native system as reflexes of /ĭj/: words like **мариѡ** < Μαρία, **стаѡи** < στάδιον, which are well-integrated into the morphological system as a fem. ja-stem and masc. jo-stem respectively, could either be reconstructed as consciously-foreign *marijĀ, *stadijъ, or as nativised *marъjĀ, *stadъjъ, but there are no occurrences of jer-spellings in these words in the OCS texts in TOROT. Other similarly-Greek words like **дѡволъ** (< διάβολος), however, do show up in OCS with jer-spellings: Supr. **дѡѡѡла**, Zogr. Luke 8 and Psal. Psalm 108 **ѡѡѡѡѡѡ** {dĭěvolŭ}, which (alongside the modern Macedonian *ѡѡѡѡ* with the reflex of *ĭj produced by the Macedonian so-called ‘new jotation’ of /d/ after the fallen jer brought it into contact with /j/) clearly suggest an early adaptation of this foreign /ij/-group to native /ĭj/. Old Russian texts even show spellings of **мариѡ** suggestive of full nativisation: Laurentian Primary Chronicle **мѡрьѡ**, **мѡрьѡю**, Zadonshchina **мѡрьѡ**, **мѡрьѡ**, as well as First Novgorod Chronicle gen. sg. **ѡѡѡѡѡѡ** (jo-stem **ѡѡѡѡѡѡ** < Βασιλειος).

1.4.3 Word-initial *j_B-/*ji-/*i-

12 Marianus and Psalterium Sinaiticum, on the other hand, frequently show a Russian-style /ej/ reflex of strong tense *bj: Psal. **vrabēi** <*vorbъjъ, **plūtei** <*plъtъjъ, **mīnei** <*mъnъjъ; Mar. **zapovēdei** <*zapovēdbъjъ, **udarei** <*udařъjъ, **gvozdeinūe** <*gvozdbъjъny-jě. Psal. (Psalm 21) even has a past act. part. Nsg. masc. def. form **strūgoi** <*jъstrgъjъ that suggests an /oj/ reflex of *bjъ.

I make an exception for certain forms of the personal-pronoun *jъ, however, and write *jimъ, *jima, *jixъ *jimъ and *jimi for the masc/nt. instr. sg. and dat./instr. dual/pl., because Czech here has *jim jich jimi*.

Prefixed forms like *do-jyti 'to come, arrive' for morphological reasons have to be distinguished from the class 4 verb *dojiti/dojiši/dojimъ etc. 'to breastfeed' (and its derived noun *dojidlika), a difference which is reflected in the modern Ukrainian *diūmu* (<*dojyti with compensatorily-lengthened /o/ > /i/) vs *doimu*. Thus /i/ can follow /j/ when the former is part of a morpheme which just happens to be stuck onto a /j/-ending stem: I similarly allow words like *šujika (шунца) and *vojinyъ 'warrior' (воинъ, as opposed to *vojnъ, the gen. pl. of *voyna), or the loc. sg./pl. desinences of any jo-stem noun whose stem ends on /j/, e.g. Psal. **жрѣбїи** {žrěbŭ} <*žerbyj.

No Glagolitic text makes any effort to distinguish /je/ (after vowels or word-initially) from post-consonantal /e/, writing both with <ѣ>, unlike the situation with the reflexes of *jɛ vs *ɛ, where in Zogr. and Mar. and partially in Assem. (Velcheva 1981: 168) the full front-nasal digraph <ѣѣ> is reserved for *jɛ, while just the second 'nasalising component' <ѣ> is used for post-consantal *ɛ, e.g. Mar. 3rd pl. aorist **ѣѣѣ** {ɛsN} <*jɛsɛ, as opposed to KF **ѣѣѣѣ** {priɛti} <*priɛti vs **ѣѣѣѣѣ** {vŭzeli} <*vŭzeli¹⁶.

15 Spellings like Zogr. Mark 13:3 “**ᲞᲣᲠᲐᲗ. Ვ Მ᲏ᲑᲙᲟ. Ვ Ბ᲏ᲢᲦᲩ.**” {petrū. ī ēkovū. ī gannū} “Peter and Jacob and John” would suggest that this initial *-i- can get dropped after an /i/ of a preceding word, but whether this points to a dropping of the non-native *-i-, simple deletion of a double /i i/ (haplology), or a native-like reflex of a weak-jer /*i *jɔ̌jÆkovŋ/ > /i jakov/, is not really knowable, so indexing such words with a markedly foreign initial *-ij- group is again the best way of allowing such difficult cases to be investigated.

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17 The leftover 14 are things like 1st. pres. dual. **ѣмаѣѣ** which Eckhoff's corpus wrongly lemmatises as **имаѣи** instead of **имѣѣи**, and which thus get reconstructed as ***jemlěvę** instead of ***jъmavę**. At the time of writing only 3227/6862 Suprasliensis lemmas have been reconstructed, but those 3227 cover 89713/99194, or 90.4%, of the words.

Therefore even though *sr > *str and *zr > *zdr appear to be simply voiced and unvoiced variants of the same sound change, the practical effects are very different because the former is, from the

For this reason I don't include /zdr/ <*zr at prefix or preposition-boundaries in my LCS system, so that investigators can see for themselves the extent of each text's adherence to the expected phonological development vs restoration of /zr/ under morphological pressure.

1.6 Morphological innovations that scupper LCS reconstruction

An example of such morphological change contingent upon structural phonological change, leading to forms which preclude any direct LCS-stage reconstruction, is the replacement of i-stem endings with those of the corresponding jo- or j̄a-stems, in nouns whose stems end on labials or the subset of LCS dental consonants which lack palatal counterparts, viz. /d t s z/²⁰. Evidence for such a change is furnished by the Old Russian masc gen./acc. form **ТАТА** from the 1229 Treaty between Smolensk, Riga and Gotland (Version A). LCS *taty is a masc. i-stem noun with genitive *tati, as it still appears in the Suprasliensis translation of John Chrysostom's Homily for Holy Thursday (...**ТО КАЖЕТЪ ВЛАДЫКЪЗІ ЧЛОВѢКОЛЮБІЕ ІАКО ПРѢДАННИКА РАЗВОЙНИКА ТАТИ**...), but in the dialect underlying the 1229 Treaty the rise of phonemically palatalised /t'/ after the Jer Shift means that the stem (and the nom. sg. **ТАТЬ** /tat'/) of this noun now ends on the same class of "soft" consonants as original

20. In some dialects (notably East Slavic) the PV3 reflexes *š and *ž became palatalised counterparts to plain /s z/, i.e. /s' z'/. and merged with the /s' z'/ that developed from LCS *s,z before front-vowels, but in most OCS they seem to have just hardened to /s, z/: searching my database for the sequence *ьхъ, for example, turns up exclusively <ѡѣ> spellings in Marianus, just one <ѡѣ> in Zogr., and exclusively <ѡѣ> in Suprasliensis, with only Assem. and Psal. containing a significant minority of <ѡѣ> spellings.

LCS / $\bar{E}$ /, though, by definition can only occur after LCS palatal consonants (see above), so a reconstruction *tat $\bar{E}$ is just nonsensical. In the case of the dat. sg. /u/-desinence (which isn't attested in our Treaty but exists in modern Russian *mamjo*), we don't even have an LCS archiphoneme available to signal a preceding soft-consonant; there's simply no way of getting from LCS *tatu to Russian /tat'u/, because such a form was only made possible by the rise of phonemic /t'/, so our ability to index it with our LCS system is gone.

Forms like **T A T A**, then, though they frustrate our goal of reconstructing entire texts, do provide us some objective measure of ‘linguistic distance’ between stages of a language, because their existence presupposes at least one intervening stage where the structure of the phonological system has changed enough from our LCS stage to have caused/allowed restructuring of the morphological system.

2.1 The existing TOROT data

21 Russian feminine i-stems like *весь* (<**вьсь*, ‘village’) do not fall together with ja-stems in the way the masculines like **звьѣ* fall together with jo-stems, but they do all still take the /am, ax, ami/ endings in the dat., loc. and instr. pl., e.g. *вѣсям*, which contain the same LCS ja-stem *-*Ā̃*-vocalism which can only occur after LCS palatals, meaning they too end up totally unreconstructable due to an illegal ***sĀ̃* sequence.

23 Except the numerous gen. sg. *ꙗꙗꙗꙗ* and dat. sg. *ꙗꙗꙗꙗꙗ* for the i-stem **gospodь*, but this word seems to be an isolated special case, because it bafflingly turns up even in early Glagolitic OCS with endings like *ꙗꙗꙗ* {*gvi*} (ju-stem dat. sg. *-*evi*, cf. Supr. *ꙗꙗꙗꙗꙗꙗ*), *ꙗꙗꙗ* {*gŭi*} (jo-stem dat. sg. *-*u*), and *ꙗꙗꙗ* {*gě*} (jo-stem gen. sg. *-*Ē*). See Van Wijk (1929).

```
<token id="3589172" form="ВЪЗВѢШТЖ" citation-part="70.17"
lemma="ВЪЗВѢСТИТИ" part-of-speech="V-" morphology="1spia----i"
relation="pred" presentation-after=" " />
```

Figure 1: TOROT annotation for *Psal. Sin. Psalm 70* ВЪЗВѢШТЖ in XML format

TOROT token XML-tags include various attributes, but for the Autoreconstructor all that's needed are *form*, *lemma*, *part-of-speech*, and *morphology*. The *form* attribute is used to check for morphological variations/innovations, to ensure that what gets produced is the direct phonological ancestor of the actually-occurring word (see below for more about this deviance-detection). The *lemma* and *part-of-speech* attributes, when concatenated, serve as a unique key linking each word to its lemma²⁴ and thus to its LCS reconstruction and inflection-class information²⁵. Finally the *morphology* attribute consists of a 10-character string to hold values for the 10 morphological-features used by TOROT (and the wider PROIEL corpora). Not all features are relevant for all words, in which case a dash '-' is used as a placeholder.

A detailed explanation of each feature can be found in Section 6 of Eckhoff et al. (2018: 41), but here it suffices to say that in this example the tag "1spia----i" is telling us that ВЪЗВѢШТЖ {vŭzvěštq} is 1st person, singular, present-tense, indicative-mood, active-voice, has no gender, case, degree, or strength features, and is inflectable rather than non-inflecting.

Of importance here is the *present* tense tagging, even though ВЪЗВѢСТИТИ (even in OCS) can be taken as a perfective verb, opposed to its imperfective counterpart ВЪЗВѢЩАТИ , and thus has future-tense meaning in its non-past indicative forms (and the Greek Septuagint here has $\alpha\nu\alpha\gamma\gamma\epsilon\lambda\omega\ \tau\acute{\alpha}\ \theta\alpha\nu\mu\acute{\alpha}\sigma\iota\acute{\alpha}\ \sigma\omicron\upsilon$, with a morphologically future-formed $\alpha\nu\alpha\gamma\gamma\epsilon\lambda\omega$ 'I will proclaim' from $\alpha\nu\alpha\gamma\gamma\epsilon\lambda\lambda\omega$). The tagging thus follows the *inflectional*-morphology of ВЪЗВѢШТЖ {vŭzvěštq}, rather than the future-meaning which is carried by the *derivational*-morphology. This is important because the Autoreconstructor works by *inflecting* LCS lemmas according to those morphology-tags; the annotation gives no information about its derivational-morphology or derivational relationship to its imperfective (and thus 1sg. *present* tense) counterpart ВЪЗВѢЩАЖ {vŭzvěštaq}, since that is annotated with a separate lemma.

An even clearer example where this *form over function* annotation helps us is with the accusatives of animate nouns: it's well known that a type of so-called differential-object-marking is beginning to take hold in Early Slavic, whereby syntactic accusatives use genitive endings to varying extents as a way of encoding the semantic ensouledness (and possibly definiteness) of the noun (the details aren't important; for a recent thorough diachronic treatment of the topic see Eckhoff 2022), e.g. Supr. РАЗБОЙНИКА ВЪ ПОРОДѢ ВЪВЕДЕ 'he brought the robber into paradise'. If these were all tagged as accusatives, the Autoreconstructor would produce the wrong ancestor to the text-form, i.e. *orzbojnik-ъ, because the semantic information needed to decide whether this differential-object-marking is needed is not available. Happily though all such animate-accusatives are marked as genitive in TOROT: the morphology-tag for РАЗБОЙНИКА above is "-s---mg--i", so the Autoreconstructor produces *orzbojnik-a.

2.2 Computerised LCS inflectional-morphology and the Autoreconstructor

The Autoreconstructor reads the morphology-tag character-by-character and numerifies each field, and from that it computes a number which corresponds to the row of the inflection-table where the endings for that particular word's inflection-class are stored. For verbs this will be a number

24 Identical lemmas with the *same* part-of-speech tag, such as ВЕСТИ 'to lead' <*ved-ti and ВЕСТИ 'to drive' <*vez-ti, both of which have 'V-' for verb, are differentiated by appending #2 etc. to the extra homomorphs, i.e. ВЕСТИ vs. $\text{ВЕСТИ\#2}$.

25 The spreadsheet containing my reconstructions and inflection-class annotations for the TOROT OCS lemmas can be downloaded from https://github.com/12401453/torot_2023/blob/main/lemma_lists/chu_lemmas_master.xlsx

between 1 and 44 (9 person/number combinations for each of present, aorist, imperfect, and imperative, plus 8 non-finite forms, $9 \cdot 4 + 8$), and for nominals between 1 and 63 (7 cases * 3 numbers * 3 genders). A version of the function which actually implements this tag-reading process can be seen in the `OcsServer::numerifyMorphTag()` function here:

https://github.com/12401453/ocs_server/blob/main/OcsServer.cpp#L2714.²⁶

Currently each inflection-class has up to three tables associated with it, the first of which is full (i.e. contains 44 or 63 entries) and holds the ‘basic’ or ‘correct’ (from the LCS perspective) endings. The second and third tables are ‘sparse’, in that they only contain entries for those parts of the paradigm where we expect to encounter alternative forms, with those I consider ‘deviant’ or ‘innovated’ held in table 2, and ‘alternative’ but still ‘correct’ endings (i.e. LCS allomorphs) in table 3. An example of some of the endings in the three tables for basic class 1 verbs like **rehi* is given below:



```
inner_map v_11_c0 = {
{1, "q"},
{2, "eši"},
{3, "etQ"},
{4, "evě"},
{5, "eta"},
{6, "ete"},
{7, "emъ"},
{8, "ete"},
{9, "qtQ"},
{10, "ъ"},
{11, "e"},
{12, "e"},
{13, "ově"},
{14, "eta"},
{15, "ete"},
{16, "omъ"},
{17, "ete"},
{18, "q"},
};

inner_map v_11_c1 = {
{6, "eta"},
{10, "oxъ"},
{13, "oxově"},
{14, "osta"},
{15, "oste"},
{16, "oxomъ"},
{17, "oste"},
{18, "ošę"},
{24, "ěašeta"},
{37, "qtj"},
};

inner_map v_11_c2 = {
{10, "sъ"},
{13, "sově"},
{14, "sta"},
{15, "ste"},
{16, "somъ"},
{17, "ste"},
{18, "se"},
};
```

Figure 2: Inflection-endings for class 1 verbs like **rehi*, **greti* stored using `C++ std::unordered_map<int, std::string>`

Here the ‘deviant’ endings consist mostly of the ‘extended S-’, or **-ox-* aorists (e.g. Assem. *ꙗꙗѣꙗꙗꙗꙗ* {narekošN}), while the third ‘alternative’ table contains the primary S-aorist endings (e.g. Assem. *ꙗꙗꙗꙗꙗꙗꙗ* {pogrěsN} <*pogrěb-se). Clearly those primary S-aorist endings just being stuck onto a stem like **greb-* would not produce the correct LCS form, so for certain inflection-classes there are post-processing functions which do things like replace all “ebsę” with “ěsę”, or (for stems which end on RUKI-consonants like **rek-*) all “eks” with “ěx”. The full routines for verbs can be found in https://github.com/12401453/ocs_server/blob/main/autoreconstructor/verb_cleaning.h; to a large extent this is just enforcing the Slavic consonant-cluster restrictions discussed in Section 1.5.

The inflection-tables themselves are indexed by integer-keys associated with each inflection-class, such that the ‘correct’ table’s key always ends on ‘1’, meaning that shifting to the alternative tables is a matter of simply incrementing the key by either 1 or 2, and I can keep track of whether or not a form has required a deviant or alternative ending by inspecting the final value of this key²⁷:

²⁶ Extra handling is required for forms of **byti*, since that has a separate future-paradigm (and future-participle) which TOROT does actually specify separately with an ‘f’ value for the *tense* feature, as well as two variant imperfect sets (3sg. **bě* vs. **běaše*, which aren’t tagged, but which I detect), and a ‘conditional’ **bimъ*, **bi*, **bq* etc. Participles require an extra step to read their nominal-features and add their adjective-like endings.

²⁷ This is bad design and prevents me from easily handling more than one type of morphological innovation per class: for the *masc_jo* class nom. pl. I have the *i*-stem **-ъje* ending in the ‘deviances’ table (as found in e.g. Supr. *стражѣ* <*storž-ъje ‘guards’), which leaves no room for forms like Supr. *зноѣе* <*znoj-eve, Psal. *ꙗꙗꙗꙗꙗꙗ* {zmieve} <*zmъj-eve, which have the **-eve* ending from the *ju*-stems.

сѣти					Random	
☐ Corpus forms						
	Present	Aorist	Imperfect	Imperative	Participles	
1st sg.	сѣтърѣ	сѣтърѣхъ	сѣтърѣахъ	сѣтърѣмь	PRAP ¹	сѣтърѣи
2nd sg.	сѣтърѣши	сѣтърѣ сѣтърѣ	сѣтърѣаше	сѣтъри	PRAP ²	сѣтърѣциѣ
3rd sg.	сѣтърѣтъ	сѣтърѣ сѣтърѣ	сѣтърѣаше	сѣтъри	PAP	сѣтърѣ
1st du.	сѣтърѣвѣ	сѣтърѣхѣвѣ	сѣтърѣахѣвѣ	сѣтърѣвѣ	L-Part.	сѣтърѣаѣ
2nd du.	сѣтърѣта	сѣтърѣта	сѣтърѣашета	сѣтърѣта	PPP	сѣтърѣтъ сѣтърѣиѣ
3rd du.	сѣтърѣте сѣтърѣта	сѣтърѣте сѣтърѣта	сѣтърѣашете сѣтърѣашета	сѣтърѣте	PrPP	сѣтърѣомѣ
1st pl.	сѣтърѣмъ	сѣтърѣхомъ	сѣтърѣахомъ	сѣтърѣмъ	Infinitive	сѣтърѣти
2nd pl.	сѣтърѣте	сѣтърѣте	сѣтърѣашете	сѣтърѣте	Supine	сѣтърѣтъ
3rd pl.	сѣтърѣтъ	сѣтърѣ сѣтърѣаѣ	сѣтърѣахъ	сѣтърѣ		

Figure 4: Dynamically-generated paradigm for the OCS verb сѣтърѣти, based on the Autoreconstructor's computerised LCS inflectional-morphology

The accuracy of the generated-forms can then be gauged by comparing them to tables populated only by forms which actually occur in Eckhoff's corpus-texts, using the 'Corpus-forms' switch:

сѣти					Random	
☒ Corpus forms						
	Present	Aorist	Imperfect	Imperative	Participles	
1st sg.					PRAP ¹	
2nd sg.					PRAP ²	
3rd sg.	сѣтърѣтъ сѣтърѣтъ	сѣтърѣ сѣтърѣ			PAP	сѣтърѣ сѣтърѣиѣ
1st du.					L-Part.	сѣтърѣаѣ
2nd du.					PPP	сѣтърѣениѣ
3rd du.					PrPP	
1st pl.				сѣтърѣмъ	Infinitive	
2nd pl.					Supine	сѣтърѣтъ
3rd pl.		сѣтърѣаѣ				

Figure 5: Forms of the same сѣтърѣти verb as they actually occur in Eckhoff's (Cyrillicised) TOROT corpus of Church Slavic texts

An advantage of basing computer-generated paradigms on converted LCS (rather than directly on normalised OCS) is that it allows for the possibility of dialect or manuscript-specific conversions: e.g. Kiev Folia-flavoured paradigms that convert **h*, **ĥ* to <ц, з> and use <ѣ> for *all* **Ē*, or a Marianus-specific conversion that frequently shows vocalised strong-jers, or even an artificial pre-Jer Shift form of Old Russian that shows the denasalisation of **ę* and its merger with **Ē* via stochastic mixing up of <ѡ, ѡ> and <ѡ>:

	Sing.	Dual	Plural
Nom.	КЪНІАЗЬ КЪНАГЪ	КЪНАЗІА	КЪНАЗІИ
Acc.	КЪНАЗЬ КЪНАГЪ	КЪНАЗА	КЪНАЗѢ КЪНАГЪІ
Gen.	КЪНАЗІА	КЪНАЗЮ	КЪНАЗЬ КЪНАГЪ
Dat.	КЪНІАЗЮ	КЪНІАЗЕМА КЪНАГОМА	КЪНАЗЕМЪ КЪНАГОМЪ
Loc.	КЪНІАЗІИ КЪНІАЗѢ	КЪНАЗЮ	КЪНІАЗІИХЪ КЪНАЗѢХЪ
Instr.	КЪНАЗЬМЬ КЪНАГЪМЬ	КЪНАЗЕМА КЪНАГОМА	КЪНІАЗІИ КЪНАГЪІ
Voc.	КЪНІАЖЕ	КЪНІАЗА	КЪНІАЗІИ

Figure 6: Dynamically-generated paradigm of **kъnędъ* but with an Old Russian rather than OCS conversion

	Sing.	Dual	Plural
Nom.	КЪНАСЬ КЪНАГЪ	КЪНАСА	КЪНАСИ
Acc.	КЪНАСЬ КЪНАГЪ	КЪНАСА	КЪНАСА КЪНАГЪІ
Gen.	КЪНАСА	КЪНАСОУ	КЪНАСЬ КЪНАГЪ
Dat.	КЪНАСОУ	КЪНАСЕМА КЪНАГОМА	КЪНАСЕМЪ КЪНАГОМЪ
Loc.	КЪНАСИ КЪНАСѢ	КЪНАСОУ	КЪНАСИХЪ КЪНАСѢХЪ
Instr.	КЪНАСЕМЬ КЪНАГОМЬ	КЪНАСЕМА КЪНАГОМА	КЪНАСИ КЪНАГЪІ
Voc.	КЪНАЖЕ	КЪНАСА	КЪНАСИ

Figure 7: **kъnędъ* with the canonical OCS conversion for comparison

2.3 Detecting and dealing with morphological innovations

217066	приведоша	3paia----i	privedošę	privedo
217101	въздѣхнѣвъ	-supamn-si	vъzdxnъvъ	vъzdxъ
217112	разврѣзосте	3daia----i	orzvrъzoste	orzvrъzete
217261	бѣгивѣ	-supamn-si	bolgoslovivъ	bolgoslovilъ
217266	ѣша	3paia----i	jĕšę	jĕšę
217272	възаша	3paia----i	vъzęšę	vъzęšę
217302	начаша	3paia----i	načęšę	načęšę
217316	въздѣхнѣвъ	-supamn-si	vъzdxnъvъ	vъzdxъ
217455	приведоша	3paia----i	privedošę	privedo
217600	начать	3saia----i	načęť	načę
217605	сѣноу	-s---md--i	synu	synovi
217635	начать	3saia----i	načęť	načę
217787	поѣтъ	3saia----i	pojęť	poję
217832	моѣомѣ	-s---mi--i	mosijomъ	mosijemъ
217856	моѣови	-s---md--i	mosijovi	mosiju
217869	быс	3saia----i	bystъ	by
217960	сѣнѣ	-s---ml--i	synę	synu
218067	невѣрьнѣ	-s---mvpsi	nevęrъnъ	nevęrъne
218108	быс	3saia----i	bystъ	by
218173	нѣмѣ	-s---mvpwi	nęmъjъ	nęmęjъ
218175	глоухѣ	-s---mvpwi	gluxъjъ	glušejъ
218198	быстъ	3saia----i	bystъ	by
218206	оумрѣтъ	3saia----i	umertъ	umer
218388	ѣмени	-s---nl--i	jъmeni	jъmene
218419	ѣмени	-s---nl--i	jъmeni	jъmene
218561	окомѣ	-s---ni--i	okomъ	očesъmъ
218648	моѣю	-s---md--i	mōžu	mōžęvi
218688	моѣжа	-s---mg--i	mōžĕ	mōžu
218755	прѣлѣубѣ	-s---fa--i	perluby	perlubъvъ
218762	поустивѣши	-supafn-si	pustivъši	pušĕšĕi

Figure 8: Auto-detected and -reconstructed morphological deviances from a small part of the Book of Mark in Codex Zographensis

The screenshot above shows some raw data from my autoreconstructed SQLite database of the TOROT OCS texts; in this case it's forms from Zographensis (around Mark 7 to Mark 10) where the Autoreconstructor has detected morphological innovations. The fourth column shows what the Autoreconstructor thinks is the direct phonological ancestor to the text-form, but the ancestor of the 'original', 'correct', or 'default' morphological form is also generated and stored in the fifth column, so that such cases of innovation can be easily searched-for and counted (since non-innovated forms have NULL values in this column).

Types of innovation detected here include:

- extended S-aorists of class 1 verbs: 3rd pl. **приведоша** vs. **приведѣ**, 3rd dual **разврѣзосте** vs. ***разврѣзете**²⁹
- unetymological extension of the RUKI-rule-produced *š in 3rd pl. primary sigmatic aorists: **ѣша** vs. **ѣса** <*jĕd-s-ę, **възаша** vs. **възаса** <*vъzym-s-ę, **начаша** vs. **начаса** <*načъn-s-ę (neither *d, *m, nor *n have ever been RUKI sounds)
- extension of the *-nq- suffix to the past. act. part. of class 2 verbs like **въздѣхнѣти**: **въздѣхнѣвъ** (cf. Mar. **ⲁⲩⲩⲧⲏⲩⲱⲩⲥ** {usūxūšq} from **ⲟⲩⲥⲧⲏⲩⲱⲩⲥ**)

29 Koch (1990: 293) lists only sigmatic aorists as possibilities for the *-verz-/*-vřz- stem verbs, and it seems that outside of the 3rd sg. (e.g. Psal. **ⲕⲁⲩⲟⲩⲱⲩⲥⲟⲩⲥ** {razvrže}, Zogr. **ⲟⲩⲱⲩⲥⲟⲩⲥ** {otvrže}) no root-aorists are attested in any Slavic text, so maybe I am wrong to set up asigmatic root-aorists like 3rd dual *-vřzete as a possibility alongside primary sigmatic *-verste (e.g. Mar. Mark 7 **ⲕⲁⲩⲟⲩⲱⲩⲥⲟⲩⲥ** {razvrže}). My justification is that the *-verg-/*-vřg- stem verbs do attest such root-aorists, e.g. 3rd pl. **ⲟⲩⲱⲩⲥⲟⲩⲥⲱⲩⲥ** {otvřęq} in Psal. Psalm 77, and I don't see what, apart from the nature of the final stem-consonant (obstruent vs. continuant), could be grounds for classifying these two verbs differently.

- addition of the *-tQ suffix from the 3rd sg. pres. (where Q stands for NSl. *ъ vs SSL. *ѣ, see fn. 5 above) to 3rd sg. aorist forms: *НАУАТЪ, ПОИАТЪ, ОУМРѢТЪ*
- original u/ju-stem nouns taking o/jo-stem endings: dat. sg. *ѣноу, мѣжю*; loc. sg. *ѣнѣ*, gen. sg. *мѣжа*
- past act. part. of class 4 verbs using the suffix *-ivъ rather than *-jъ: *бгвивѣ, поуcтивѣши* (cf. Mar. Mark 10 *пуcтѣи* {puštŭši} <*pust-jъši)

Deciding upon the “correct” morphological endings for an unattested language inevitably entails some uncertainty and controversy: for instance, *-ox- aorists occur in both OCS and Old Russian, so why do I consider them LCS deviances? Basically because they **never**³⁰ occur in Marianus, which would be quite improbable if they were a discarded archaism (especially given their ubiquity in the closely related Zographensis). Additionally, West Slavic uses *-ex- here (e.g. Old Czech 1sg. *nesech* <*nes-ex-ъ), suggesting that the extended S-aorists were innovated independently in the post-LCS dialects (Meillet 1965: 256).

In other cases we are dealing with hodge-podge paradigms which are only ever attested with endings from multiple older classes, and sometimes dialectal or orthographic features of the manuscripts can make it difficult to distinguish potential LCS ancestors of certain endings. For instance, I have a *masc_tel* class used for agent-nouns like OCS *дѣлатѣль* (i.e. Diels 1963: 166), which in the sg. and dual. behave exactly like masc jo-stems, but which in the gen. and instr. plural are attested also with basically consonant-stem endings on a hard *-tel- stem, e.g. Zogr. *ѣтѣлѣ* {tŋžatelŭ} <*tęžĕtelŭ, Supr. *свѣтитѣлѣ* <*svĕtitylŭ. The nom. pl. appears also to take a consonant-stem *-e ending, but as Diels (op. cit.) points out, the spellings in Zogr. and Supr. (where use of the palatalisation-diacritic <^> to denote LCS *ĭ is consistent enough to suggest a real phonemic /ĭ/ in the underlying dialects) like *мѣтитѣлѣ*³¹ mean we can’t follow Meillet (1965: 426) in setting up *-tele as the ‘correct’ ending, because spellings like *ѣтѣлѣ* {žntele} in Mar. could just as easily descend from *žetelē as from *žetele. Absent a manuscript which both consistently marks *ĭ and doesn’t use such a mark in this nom. pl. desinence, there’s no hard evidence of this *-tele ending ever actually existing. I thus use *-tele in the ‘correct’ table, and the jo-stem *-telŭ in the ‘deviances’ table³².

2.4 Overcoming poor lemmatisation practice

Sometimes Eckhoff’s lemmatisation is too coarse-grained, in that forms which clearly descend from distinct doublets are subsumed under one lemma. To demonstrate just one example of how the Autoreconstructor deals with this sort of problem, take the numeral *ѣдинъ* <*jedīnъ, which in the earliest OCS has straightforward hard pronominal endings and which therefore goes in my *pron_hard* class alongside demonstratives like *онъ. There is, however, what must be viewed as an LCS doublet *jedŭnъ³³, which gives e.g. Serbian *jedan* and the modern Russian fem. nt. and

30 Having autoreconstructed all of Marianus I can verify this (admittedly already well-established) fact far more quickly and easily than was possible just with Eckhoff’s morphology and part-of-speech annotation-information: using TOROT the smallest net you could cast would be one that caught all aorist-tense verbs, whereas I can just search for reflexes of *oxъ, *oxov, *oxom, *oŝe, *osta\$, and *oste\$ (the latter two using ‘regular-expression’ mode and \$ to specify end-of-word), which for Mar. turns up nothing except the three occurrences of *оста* {osta}.

31 Searching my database for *telē\$ returns only a single *вѣстѣлѣ* in Supr. without the diacritic; everything in Zogr. has it.

32 Diels mentions only Psal. Psalm 26 *сѣвѣдѣлѣ* {sŭvĕdĕtelŭ}, but the Autoreconstructor is intended for use with other early texts beyond canonical OCS which might also contain this type of assimilation to the jo-stems.

33 This despite Meillet’s (1965: 144) incoherent speculation about the /i/ in *jedīnъ resulting from a phonetic development of strong *ъ, analogous to what happens in the vicinity of *j, because “or il s’agit d’un composé dont le second élément est *jīnŭ”. While this interpretation of *jedīnъ’s derivation could be true (and according to Derksen’s (2008: 212) etymology of the *jŭnъ pronoun, its *j is prothetic, meaning it wouldn’t develop if already attached to a stem ending in *d), the idea that a synchronic *-dŭnъ by any purely phonological means could become /-din/, let alone early enough to completely displace by analogy the oblique forms in the earliest OCS where *ѣдинъ* {edin-} spellings are overwhelming, is ludicrous and contradicted by all the philological evidence.

oblique-case forms *одна, одного* etc., and which is used for the majority of non masc. nom./acc. sg. forms of the pronoun in Suprasliensis³⁴, e.g. *ѦД'НОМОУ*. Notwithstanding Eckhoff's habit of lemmatising these with *ѦДИНЪ* instead of *ѦДЫНЪ*, it's trivial for the Autoreconstructor to check the form (which has in a previous step already been aggressively normalised to get rid of morphologically-irrelevant variation) for a <ДИН> sequence, which would never occur in the inflectional-ending and so always point to a *jedin-descended stem, and then replace our autoreconstruction with *jedьн- if such isn't found:

```
// check for *jedьн-
if (lemma_ref.lemma_id == compileTimeHashString("РѦДИНЪ") || lemma_ref.lemma_id == compileTimeHashString("МѦДИНЪ"))
{
    if (Sniff(cyr_id, "дін", 20) == false)
    {
        stem = "jedьн";
    }
}
```

Figure 9: Part of the Autoreconstructor's code which checks for reflexes of *jedьн- that TOROT has mistakenly lemmatised under *ѦДИНЪ*

In the long-term TOROT itself should fix such lemmatisation problems, but in the meantime checks like the above are computationally extremely cheap and prevent great swathes of the texts from being wrongly autoreconstructed.

3. Benefits and potential for future studies

We have already seen a few examples of minor questions where recourse to an LCS reconstructed layer makes exhaustive investigation of texts trivial; e.g. the reality of a difference between word-initial /je-/ and (foreign) /e-/ in the Codex Suprasliensis was shown by demonstrating statistically significant discrimination in the use of the <Ѧ> letter (p.9-10). Another example, as hinted at in fn. 20, concerns the fate of the PV3 reflexes *ś and *ž, specifically whether they harden and merge with plain /s z/, or become palatalised counterparts to plain /s z/ (/s' z'/) as in East Slavic, which is a very important indicator of the extent to which a phonological system is developing Russian-style phonemic secondary palatalisation of consonants.

By far the largest benefit though of the type of annotation proposed here would be felt in questions of a more systematic and holistic nature, where cross-referencing of multiple orthographic (and potentially also morphological) features must be undertaken to corroborate hypotheses. The extremely important issue of the development of phonemic secondary consonant-palatalisation before various LCS front-vowels, for example, which is a critical isogloss in both Bulgaro-Macedonian and East Slavic, is just such a problem, but since there are no separate letters in either Glagolitic or Cyrillic for palatalised consonants, its existence can only be deduced indirectly by combining evidence from a range of other indicators. In addition to changes in nominal morphology mentioned in Section 1.6 (the adoption by *i-stems of *jo/*ja-stem endings), Townsend & Janda (1996: 107-8) bring our attention to a fascinating inverse-relationship in the Slavic dialects between the extent of phonemic tonality-distinctions (hard/soft opposition) in consonants on the one hand, and the longer maintenance of pitch-distinctions and distinctive vowel-length on the other. This is linked to earlier loss of jers in central vs. peripheral areas, with e.g. far southwestern Ukrainian being the most central part of East Slavic and thus most likely to have either lost or not developed secondary consonant palatalisation, and to have longer maintained distinctive vowel-length (on the East Slavic material specifically see also Trubetzkoy 1924).

34 According to my database there are 145 reflexes of *jedьн- in this category vs 46 from *jedin-, though all 107 reflexes of the masc. nom./acc. sg. are from *jedinъ. Interestingly the Uspenskij Sbornik (at least the parts included in TOROT), in contrast to later Russian, appears to completely lack *jedьн-, though all except the single *ѦДННОН* are spelt with the OCS <Ѧ/ѥ> reflex of initial *je-.

An LCS index makes testing for such correlations much easier, not only because of the speed and ease with which things like the Jer Shift can be investigated, where there are simply so many data-points that exhaustive manual investigations would be a significant undertaking, but also because relevant changes often involve unpredictable or zero-reflexes, which it is simply not possible to reliably search for via surface-forms. The tolerance of bimoraic syllables and distinctive vowel-length, for instance, may be probeable by looking at things like loss of intervocalic /j/ and subsequent vowel-contractions (in e.g. the 3sg. present ending of class 5.4 verbs *-ajetQ, which in some OCS texts is reflected as <-ааѣ>, or long-adjective endings like *o-je, *a-jĒ, which are the source of Ukrainian neuter -e and fem. -a endings, opposed to Russian uncontracted -oe and -ая), but one cannot search the surface-texts for the *absence* of something. One can however search the LCS index for sequences like *ajĒ or *ějĒ to see whether a certain text has more or fewer spellings indicative of contraction.

While detailed studies into individual manuscripts' treatment of the jers were conducted more than a century ago by figures like Jagić (1876, Zogr.) and Leskien (1905, Zogr./Mar.), they contain little attempt to integrate their findings into something we would recognise as a coherent phonological description. Jagić's study, which is the origin of the so-called 'Jer Umlaut' theory (the view that weak jers in Zographensis especially were fronted/backed by the vowel in the following syllable, p.17-18), in particular is full of incorrect etymologies³⁵, and although Leskien's is more systematic and free of etymological errors, the fact that he was writing before our concept of the phoneme shows itself everywhere in his reasoning and in the implausibility of his explanations. His ultimate conclusion is that the apparently 'umlauted' weak-jer spellings in fact reflect the hardness/softness of the consonant-cluster brought into being by the disappearance of the weak-jer (p.347-8), and therefore that front-vowels in following syllables softened in some way the preceding consonant, i.e. *zľě > /zl'ě/ > /z'l'ě/ <ꞥꞥꞥꞥ> {zĩľ}. On p.325 though he explicitly denies "dass die Palatalisirung durch folgende weiche Vokale denselben Grad der Stärke besass wie die der altererbtten palatalen Consonanten", and so must be envisaging a rather farfetched three-way phonological opposition between e.g. /n/, /n'/, and /ń/.³⁶ Lunt (1952: 322-5) then looked again at Zographensis, and adduces data he manually gathered from "about half the manuscript" to broadly support Jagić's 'Jer Umlaut' theory, but since he only gives rough percentages and we don't have the exact data he draws them from, we have no way of verifying it ourselves.

35 E.g. p.23 “**кѡлати** halte ich für richtigere als **клати**” (*kolti), and on p.35 he assumes a jer-fronting in **зѣрѣти** based on his (also incorrect) assumption that the jer in **бѣрати** is original.

theory no conclusions whatsoever can be drawn about incorrect weak back-*jer* spellings before LCS hard consonants in OCS, because they could equally easily denote softening and then *jer*-merging as simple lack of softening and loss of the vowel.

LCS reconstructions are also useful for checking uncertain etymologies: if a text consistently spells a certain sound-group in securely-reconstructed words one way, but a different etymon which we less-assuredly reconstruct with that same sound-group is spelt differently, then that's a strong indication that our candidate-etymology is faulty. For example, the entry for Russian *бренie* in Vasmer's etymological dictionary (Фасмер 1986) mentions Sobolevskiy's suggested *bŕňje reconstruction with syllabic *ŕ instead of the more traditional *bŕňnje with a TRŦT-group, based on East Slavic **БРННѢ** and **БРННѢ** spellings in a.o. the Izbornik 1073 and the Archangel'sk Gospel 1092. Sobolevskiy's view is that Russian *бренie* is a written borrowing from Church Slavonic which reflects the OCS <liquid + jer> ordering of the letters for syllabic liquids, viz. **БРННѢ**.

38 Though note that the modern Macedonian *semne* <*zēbne[tɔ̌] 'shiver' also has an unetymological /dz/ in this root, as do many other Macedonian words with LCS *z such as *svep* <*zvěř̥.

Both the resulting annotations and the computational apparatus used to produce them have value not only for researchers, but also in their potential for creating innovative pedagogical tools for learners. The work-in-progress ocstexts.co.uk website aims to demonstrate some of these things: for example ‘normalised’ spelling of a text’s words, based on conversions from the regular autoreconstructed LCS forms, can help students learn about and cope with the significant dialectal and orthographic variation characteristic of all OCS texts (Figure 10), and the computerised LCS inflectional morphology which is used to generate the autoreconstructions of single forms can be repurposed to dynamically-generate the whole paradigm of every reconstructed lemma, resulting in comprehensive and high-quality inflection-tables (see p.16). We can even combine this paradigm-generating ability with each text-word’s TOROT morphology-tag to visually locate a specific form in its generated table, which massively helps with the learning of paradigms (Figure 11).

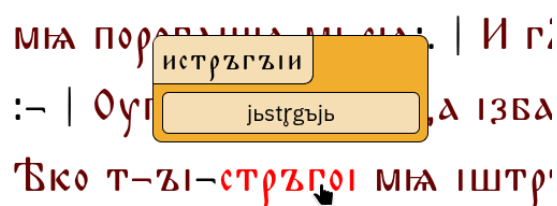


Figure 10: Tooltip showing the autoreconstructed LCS ancestor of the past active participle of the verb *истрѣгъи* in *Psalterium Sinaiticum*, along with its conversion into 'normalised' OCS

	Masc.	Sing.	Dual	Plural
Nom.	дѣнь	дѣни	дѣне	дѣнне
Acc.	дѣнь	дѣни	дѣни	дѣни
Gen.	дѣне	дѣноу	дѣноу	дѣни
Dat.	дѣни	дѣнма	дѣнма	дѣнма
Loc.	дѣни	дѣноу	дѣноу	дѣнхъ
Instr.	дѣнмъ	дѣнма	дѣнма	дѣнми
Voc.	дѣнь	дѣни	дѣне	дѣнне

Figure 11: Paradigm of the noun *дѣнь* generated after Ctrl+clicking on an instrumental-plural occurrence in the *Codex Marianus*, with the corresponding table-entry animated

In many ways the point of what I’m proposing with this article is simply to make maximally accessible and applicable for manuscript-investigators the knowledge already accumulated by the nearly two centuries of serious Slavonic scholarship, i.e. the sum of our etymological knowledge (as distilled in etymological dictionaries like Derksen (2008) and the ESSJa) combined with what we know about Common Slavonic phonology and morphology. Furnishing every word of a text with its LCS ancestor simply brings this accumulated knowledge directly ‘on top’ of the philological evidence at a granular, per-word level, so that deviations from this regular theoretical yardstick are visible, retrievable, and quantifiable.

Though important OCS and Old East Slavic texts are still lacking the sort of annotation-data needed for the method of autoreconstruction outlined here, it should be noted that recent advances in neural-network-based tagging and lemmatisation (e.g. Rabus & Besters-Dilger 2021) mean that modern taggers trained on the existing manually-annotated corpus will perform extremely well on

the remaining canonical OCS texts, which massively cuts down the time and effort required to do for, say, the Savvina Kniga what has already been done for the Marianus. As the volume and quality of training-data increases, the performance of such automatic taggers can only improve, which will open up more and more of the early Slavic written record to the sort of automatic phonological annotation described here.

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Appendix 1: Glagolitic transliteration table

Glagolitic	Latin		Commonly-used Cyrillic
ⱁ	a		А
ⱂ	b		Б
ⱃ	v		В
ⱄ	g		Г
ⱅ	d		Д
ⱆ	e		Е
ⱇ	ž		Ж
ⱈ	z		З, Ѕ
ⱉ	z		З, Ѕ
ⱊ	ī		Ї, І
ⱋ	l		І
ⱌ	i		И
ⱍ	ǵ		Ц, Ѧ, ѧ
ⱎ	k		К
ⱏ	l		Л
ⱐ	m		М
ⱑ	n		Н
ⱒ	o		О
ⱓ	p		П
ⱔ	r		Р
ⱕ	s		С
ⱖ	t		Т
ⱗ	ü		У, Ў
ⱘ	f		Ф
ⱙ	x		Х
ⱚ	ω		W
ⱛ	θ		Ѳ
ⱜ	c		Ц
ⱝ	č		Ч
ⱞ	š		Ш
ⱟ	ć		Щ
Ⱡ	ǔ		Ъ
ⱡ	ĩ		Ы
Ɫ	ě		Ѣ
Ᵽ	u		Ѹ, ѹ
Ɽ	ü		Ю
ⱦ	Q		Ѡ
ⱨ	Ö		Ѣ̋
ⱪ	ę		Ѧ̋, ѧ̋
ⱬ	N		Ѧ
Ɱ	Ń		Ѧ̋